tautological 1-form on $M = T^*Q$

$T_qQ \cong T_q^*Q$ as vect spaces

$TQ \cong T^*Q$ as bundles

coordinate free def. $\theta : M \to T^*M$ (section of $T^*M = T^*(T^*Q)$)

$m \mapsto (m, \theta_m)$

$\theta_m \in T_m^*M$

$\theta_m : T_mM \to \mathbb{R}$ linear

$\pi : (q,\alpha) \in M \mapsto q \in Q$

$d\pi_{(q,\alpha)} : T_{(q,\alpha)}M \to T_qQ$

if $m = (q, \alpha)$ where $q \in Q$, $\alpha \in T^*Q$

then we define $\theta_{(q,\alpha)}(v) = \alpha \circ d\pi_{(q,\alpha)}(v)$

$T^*(T^*Q)$

$T^*Q = (TQ)^*$

$(T^*Q)^* = TQ$

<u>in coordinates</u> if $q^1, \dots, q^n$ coordinates on Q

induce coordinates on $M = T^*Q$ as $(q^1, \dots, q^n, p_1, \dots, p_n)$

$$p_\mu(q, \underbrace{\alpha_\nu dq^\nu_q}_{\in T_q^*Q}) = \alpha_\mu$$

$$\theta_{(q, \alpha_\mu dq^\mu_q)}(v) = \alpha_\mu dq^\mu_q \left(d\pi_{(q, \alpha_\mu dq^\mu_q)}(v) \right)$$

basis is $\frac{\partial}{\partial q^\mu}, \frac{\partial}{\partial p_\mu}$

$$d\pi_{(q, \alpha_\mu dq^\mu_q)}(v) = ?$$

$v \in T_{(q,\alpha)}M$

$$v = A^\mu \frac{\partial}{\partial q^\mu}\Big|_{(q,\alpha)} + B_\mu \frac{\partial}{\partial p_\mu}\Big|_{(q,\alpha)}$$

$$d\pi_{(q, \alpha_\mu dq^\mu_q)} \left(\frac{\partial}{\partial q^\mu}\Big|_{(q,\alpha)} \right) f$$

$\to f : Q \to \mathbb{R}$

$f \circ \pi : M \to Q \to \mathbb{R}$

$$= \frac{\partial}{\partial q^\mu}\Big|_{(q,\alpha)} (f \circ \pi)$$

$f : M \to N$

$df_q : T_qM \to T_{f(q)}N$

$df_q(v) F$ $\quad F : N \to \mathbb{R}$

$\parallel$

$v(F \circ f)$

$$= \frac{\partial f}{\partial q^\mu}(q)$$

$$d\pi_{(q, \alpha_\nu dq^\nu_q)} \left(\underbrace{\frac{\partial}{\partial q^\mu}\Big|_{(q,\alpha)}}_{\in T^*_{q,\alpha}M} \right) = \underbrace{\frac{\partial}{\partial q^\mu}\Big|_q}_{\in T_q^*Q}$$

$$d\pi_{(q,\alpha_\nu dq^\nu_q)}\left(\frac{\partial}{\partial p^\mu}\Big|_{(q,\alpha)}\right) f$$

$$= \frac{\partial}{\partial p^\mu}\Big|_{(q,\alpha)} f\circ\pi \quad = 0$$

$$\theta_{(q,\alpha_\mu dq^\mu_q)}(v) = \alpha_\mu\, dq^\mu_q\left(d\pi_{(q,\alpha_\mu dq^\mu_q)}(v)\right)$$

$$v = \frac{\partial}{\partial q^\nu}\Big|_{(q,\alpha)}$$

$$\theta_{(q,\alpha_\mu dq^\mu_q)}\left(\frac{\partial}{\partial q^\nu}\Big|_{(q,\alpha)}\right) = \alpha_\mu \overbrace{dq^\mu_q\left(\frac{\partial}{\partial q^\nu}\Big|_q\right)}^{=\delta^\mu_\nu} = \alpha_\nu = p_\nu(q,\alpha)$$

$$\theta_{(q,\alpha_\mu dq^\mu_q)}\left(\frac{\partial}{\partial p^\nu}\Big|_{(q,\alpha)}\right) = 0$$

$$\left.\begin{aligned}
&\theta_{(q,\alpha)} = A_\mu\, dq^\mu_{(q,\alpha)} + B^\mu\, dp_{\mu(q,\alpha)}\\
&\theta_{(q,\alpha)}\left(\frac{\partial}{\partial q^\nu}\Big|_{(q,\alpha)}\right) = A_\nu = p_\nu(q,\alpha)\\
&\theta_{(q,\alpha)}\left(\frac{\partial}{\partial p_\nu}\Big|_{(q,\alpha)}\right) = B^\nu = 0
\end{aligned}\right\} \Rightarrow \boxed{\theta = p_\mu\, dq^\mu}$$

$$\omega = -d\theta = dq^\mu \wedge dp_\mu \qquad \text{non-degenerate}$$

<u>charge coordinates:</u> $\quad x^i = \begin{cases} q^i & \text{if } i=1,\dots,n \\ p^{i-n} & \text{if } i=n+1,\dots,2n \end{cases}$

$$(x^1,\dots,x^{2n}) = (q^1,\dots,q^n,p_1,\dots,p_n)$$

$$\omega = \sum_{i=1}^n dq^i\wedge dp_i = \sum_{i=1}^n dx^i\wedge dx^{i+n} = \underbrace{\sum_{i=1}^n\left(dx^i\otimes dx^{i+n} - dx^{i+n}\otimes dx^i\right)}$$

matrix associated to ω ?

$$\omega = \underbrace{\sum_{i,j=1}^{2n}\Omega_{ij}\, dx^i\otimes dx^j} \qquad (\Omega_{ij}) \text{ matrix}$$

$$\Omega_{ij} = \omega\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right)$$

$$\sum_{k=1}^{n}\left(dx^k \otimes dx^{k+n} - dx^{k+n} \otimes dx^k\right) = \sum_{i,j=1}^{2n}\left(\delta_{j,i+n} - \delta_{i,j+n}\right) dx^i \otimes dx^j$$

$$\Omega_{ij} = \delta_{j,i+n} - \delta_{i,j+n}$$

$$\begin{array}{c|c} & \begin{array}{cccc|cccc} j=1 & j=2 & \cdots & j=n & j=n+1 & \cdots & & j=2n \end{array} \\ \begin{array}{c} i=1 \\ i=2 \\ \\ i=n+1 \\ \vdots \\ i=2n \end{array} & \left(\begin{array}{cccc|cccc} 0 & 0 & & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 1 & 0 & 0 \\ \hline -1 & 0 & \cdots & & 0 & 0 & 0 & 0 \\ 0 & -1 & \cdots & & 0 & 0 & 0 & \cdots \end{array}\right) \end{array}$$

$$\Omega = \begin{pmatrix} 0 & \mathbb{1}_n \\ -\mathbb{1}_n & 0 \end{pmatrix}$$

$$\det\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(AD - BC)$$

$\det \Omega = 1$

if $[C,D]=0$ or $[A,B]=0$

or $[A,C]=0$ or $[B,D]=0$

dynamics $\leftrightarrow$ ODE

1st order ODE on $\mathbb{R}^n$

$$\begin{cases} \dfrac{dy}{dt} = f(y(t)) \\ y(0) = y_0 \end{cases}$$

$y(t) \in \mathbb{R}^n$

$f(y) \in \mathbb{R}^n$

note: $y : (-\varepsilon, \varepsilon) \to \mathbb{R}^n$

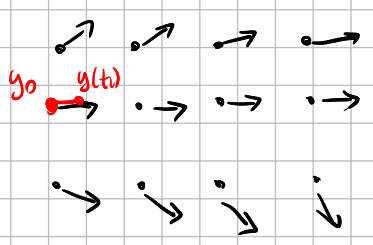
y curve in $\mathbb{R}^n$

$f : \mathbb{R}^n \to \mathbb{R}^n$ vector field

$\frac{dy}{dt}$ velocity of curve

$\to$ tangent vector to curve

graphical idea: f associates a vector at each point



1st order ODE on M

$\gamma : (-\varepsilon, \varepsilon) \to M$ smooth curve (eq. to y)

tangent vector to γ at parameter t $(\dot\gamma(t))$

$\dot\gamma(t) f = \frac{d}{ds} f(\gamma(s))\Big|_{s=t}$ as a derivation

equal to a vector field X evaluated at point $\gamma(t)$

$$\boxed{\begin{cases} \dot\gamma(t) = X\big|_{\gamma(t)} \\ \gamma(0) = \gamma_0 \end{cases}}$$

$X : M \to TM$

$X\big|_{\gamma(t)} \in T_{\gamma(t)}M$

$\dot\gamma(t) \in T_{\gamma(t)}M$

for (M, ω) symplectic we can define for each $f: M \to \mathbb{R}$ a vector field X_f s.t. $\omega(X_f, \cdot) = df$

once we choose a special function $H: M \to \mathbb{R}$ we can build X_H and the ODE

$$\begin{cases} \dot{\gamma}(t) = X_H|_{\gamma(t)} \\ \gamma(0) = \gamma_0 \end{cases}$$

Hamilton's equations

in coords if $\omega = dq^\mu \wedge dp_\mu$ the eqs are

$$\begin{cases} \dot{q}^\mu = \dfrac{\partial H}{\partial p_\mu} \\ \dot{p}_\mu = -\dfrac{\partial H}{\partial q^\mu} \\ q(0) = q_0 \\ p(0) = p_0 \end{cases}$$

<u>Poisson bracket</u> A poisson bracket on M is a bilinear map $\{\cdot,\cdot\} : \mathcal{C}^\infty(M,\mathbb{R}) \times \mathcal{C}^\infty(M,\mathbb{R}) \to \mathcal{C}^\infty(M,\mathbb{R})$ s.t.

- $\{\cdot,\cdot\}$ is a Lie bracket (anti-symmetric, Jacobi identity $\{f,\{g,h\}\} + \{g,\{h,f\}\} + \{h,\{f,g\}\} = 0$)
- satisfies Leibniz rule $\{fg,h\} = \{f,h\}g + f\{g,h\}$

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can use Poisson bracket to define hamiltonian vector fields $f \in \mathcal{C}^\infty(M,\mathbb{R})$ $X_f g = \{g,f\}$ → can do dynamics using this

---

if $f: \mathcal{C}^\infty(M,\mathbb{R})$ H is Hamiltonian and γ is a solution of $\dot{\gamma} = X_H$

we can ask how f evolves

what is $\frac{d}{dt} f(\gamma(t))$?

$$\frac{d}{dt} f(\gamma(t)) = (f \circ \gamma)'(t) = \underbrace{\dot{\gamma}(t)}_{\in T_{\gamma(t)}M \text{ (derivation)}} f = X_H|_{\gamma(t)} f = (X_H f)(\gamma(t)) = \{f, H\}(\gamma(t))$$

often people write $\frac{df}{dt} = \{f, H\}$ in Q.M: $i\hbar \frac{dA}{dt} = [A, H]$

Cool thing if (M,ω) is symplectic then $\{f,g\} = \omega(X_f, X_g)$ is Poisson

Symmetries and Noether's theorem

suppose we have a group action on (M,ω)

$$\Phi : G \times M \to M \qquad \Phi_g : M \to M \qquad \Phi_{gh} = \Phi_g \circ \Phi_h$$
$$(g,m) \mapsto \Phi_g(m) \qquad \Phi_e = id$$

if $\xi \in \mathfrak{g}$ we can look at $\Phi_{e^{t\xi}}$ and define a vector field on M associated to ξ denoted by Y_ξ

$$Y_\xi f(a) = \frac{d}{dt} f(\underbrace{\Phi_{e^{t\xi}}(a)}_{e^{t\xi} \triangleright a})\Big|_{t=0} \qquad a \in M \qquad f : M \to \mathbb{R}$$
$$Y_\xi : M \to TM$$

Y_ξ infinitesimal generator of the transformation $\Phi_{e^{t\xi}}$

the curve $\gamma : t \mapsto e^{t\xi} \triangleright a$ is a solution of $\begin{cases} \dot\gamma = Y_\xi \\ \gamma(0) = a \end{cases}$

$$[X,Y]f = X(Yf) - Y(Xf)$$
$$[X_{\underbrace{\xi_1}_{\in\mathfrak{g}}}, X_{\underbrace{\xi_2}_{\in\mathfrak{g}}}] = -X_{[\xi_1,\xi_2]}$$

what is a symmetry of M?

$\to$ a group action Φ s.t.

- Φ preserves the symplectic form $\to$ $\Phi_g^* \omega = \omega$ or equivalently $\Phi_g^*\{f_1,f_2\} = \{\Phi_g^* f_1, \Phi_g^* f_2\}$
- Φ preserves the Hamiltonian $\to \Phi_g^* H = H$
  $\to H(g \triangleright a) = H(a)$

$$\omega_a(v,w) = \omega_{\Phi_g(a)}(d\Phi_g|_a(v), d\Phi_g|_a(w))$$

if $\xi \in \mathfrak{g}$ we say that $\mu^\xi : M \to \mathbb{R}$ is a momentum map for ξ if

$d\mu^\xi = \omega(Y_\xi, \cdot)$, that is $Y_\xi = X_{\mu^\xi}$

fundamental vector field $\qquad$ Hamiltonian v. field

X_f defined by $df = \omega(X_f, \cdot)$

in general to each f we associate a v. field X_f

but we can't always reverse procedure

ex: $M = \mathbb{C}$ $\quad \omega = \frac{1}{2i} d\bar{z} \wedge dz$ $\quad H(z,\bar{z}) = |z|^2$ $\quad G = U(1)$

$$\Phi : U(1) \times \mathbb{C} \to \mathbb{C}$$
$$(e^{i\theta}, z) \mapsto e^{i\theta} z$$

$$\Phi_g^* \omega = \frac{1}{2i} d(\bar{g}\bar{z}) \wedge d(gz)$$
$$= \frac{1}{2i} \underbrace{|g|^2}_{=1} d\bar{z} \wedge dz$$

$$H(e^{i\theta} z, e^{-i\theta} \bar{z}) = |z|^2 = H(z,\bar{z})$$

momentum map? $\quad \mathfrak{g} = \underline{u}(1)$ is 1-dim

1-generator $\quad \xi = i$

$$Y_\xi f(z,\bar{z}) = \frac{d}{dt}\Big|_{t=0} f(\Phi_{e^{it}}(z,\bar{z})) = \frac{d}{dt}\Big|_{t=0} f(e^{it} z, e^{-it} \bar{z})$$

$$= \frac{\partial f}{\partial z} \frac{d}{dt}\Big|_{t=0} e^{it} z + \frac{\partial f}{\partial \bar{z}} \frac{d}{dt}\Big|_{t=0} e^{-it} \bar{z}$$

$$= iz \frac{\partial f}{\partial z} - i\bar{z} \frac{\partial f}{\partial \bar{z}} \quad \Rightarrow \quad \boxed{Y_\xi = iz \frac{\partial}{\partial z} - i\bar{z} \frac{\partial}{\partial \bar{z}}}$$

is there μ^ξ?

$$d\mu^\xi = \omega(Y_\xi, \cdot) = \frac{1}{2\cancel{i}}\left(-\cancel{i}\bar{z}\, dz - \cancel{i} z\, d\bar{z}\right) = -\frac{\bar{z}\, dz}{2} - \frac{z\, d\bar{z}}{2}$$

$$\| $$

$$\frac{\partial \mu^\xi}{\partial z} dz + \frac{\partial \mu^\xi}{\partial \bar{z}} d\bar{z} \quad \Rightarrow \quad \frac{\partial \mu^\xi}{\partial z} = -\bar{z} \qquad \frac{\partial \mu^\xi}{\partial \bar{z}} = -z$$

$$\mu^\xi = -|z|^2$$ 1 possible solution

Noether theorem Suppose $\Phi : G \times M \to M$ is a symmetry of (M, ω).

If $\xi \in \mathfrak{g}$ has a momentum map μ^ξ, then μ^ξ is conserved along solutions of Hamilton's eqs.

proof: μ^ξ conserved means that if γ is a solution of $\dot{\gamma} = X_H$ then $\mu^\xi \circ \gamma$ is constant.

check $\frac{d}{dt}(\mu^\xi \circ \gamma)(t) = \{\mu^\xi, H\}(\gamma(t)) = -\{H, \mu^\xi\}(\gamma(t))$

$\{f, g\} = X_g f \longrightarrow$

$$= -X_{\mu^\xi} H(\gamma(t))$$

$$= -Y_\xi H(\gamma(t))$$

$$= -\frac{d}{ds}\Big|_{s=0} H(e^{s\xi} \triangleright \gamma(t))$$

$$= -\frac{d}{ds}\Big|_{s=0} H(\gamma(t)) = 0$$

---

back to ex

$M = \mathbb{C}$ $\quad \omega = \frac{1}{2i} d\bar{z} \wedge dz$ $\quad H(z, \bar{z}) = |z|^2$ $\quad G = U(1)$

$\Phi : U(1) \times \mathbb{C} \to \mathbb{C}$ $\quad \Phi_g^* \omega = \omega$ $\quad \Phi_g^* H = H$

$(e^{i\theta}, z) \mapsto e^{i\theta} z$

$\mu^i : z \in \mathbb{C} \mapsto -|z|^2 \in \mathbb{R}$

Theorem tells us that μ^i is conserved

check Hamilton's eq: $X_H = -iz\frac{\partial}{\partial z} + i\bar{z}\frac{\partial}{\partial \bar{z}}$

$\dot{\gamma} = X_H \to \dot{z}\frac{\partial}{\partial z} + \dot{\bar{z}}\frac{\partial}{\partial \bar{z}} = -iz\frac{\partial}{\partial z} + i\bar{z}\frac{\partial}{\partial \bar{z}}$

$\dot{z} = -iz$

$\dot{\bar{z}} = i\bar{z}$ $\quad \to z(t) = z_0 e^{-it}$ $\quad |z(t)|^2 = |z_0|^2$ constant

$z(0) = z_0$

$\bar{z}(0) = \bar{z}_0$

ex from yesterday (Shrodinger's eq)

$M = \mathbb{C}^n$ $\quad i\hbar \frac{d}{dt} z^\mu = H^\mu{}_\nu z^\nu \to \boxed{z(t) = e^{\frac{-it}{\hbar}H} z(0)}$

$U(1)$ symmetry: $e^{i\theta} \triangleright z = e^{i\theta} z$ $\quad \mathcal{H}(\bar{z}, z) = z^* H z$

momentum map: $\mu^i : z \mapsto -\|z\|^2 \to$ conserved: Shrodinger's eq. preserves norm

$\mathcal{H}$ is also conserved $\{\mathcal{H}, \mathcal{H}\} = 0$

so $\frac{d\mathcal{H}}{dt} = \{\mathcal{H}, \mathcal{H}\} = 0$

<u>ex:</u> Harmonic oscillator $Q = \mathbb{R}$ $M = T^*Q \cong \mathbb{R}^2$

coordinates q, p $\omega = dq \wedge dp$

$H = \frac{1}{2}(q^2 + p^2)$

what are symmetries of (M, ω, H) ?

look at linear symmetries

$\Phi_g : \mathbb{R}^2 \to \mathbb{R}^2$ linear

$\begin{pmatrix} q \\ p \end{pmatrix} \mapsto A_g \begin{pmatrix} q \\ p \end{pmatrix}$ $A_g \in GL(2, \mathbb{R})$

Quantization: algebra $(C^\infty(M, \mathbb{R}), \{\cdot,\cdot\})$

$\downarrow$

algebra $(\mathrm{Lin}(\mathcal{H}, \mathcal{H}), [\cdot,\cdot])$

Canonical quantization

q, p $\{q, p\} = 1$

$\hat{q}, \hat{p}$ $[\hat{q}, \hat{p}] = i\hbar$

quantize $q^3 p^2$

$\to \hat{q}^3 \hat{p}^2 \neq \hat{q}^2 \hat{p}^2 \hat{q}$

what can A_g be?

$H(q,p) = \frac{1}{2}(q \ p)\begin{pmatrix} q \\ p \end{pmatrix}$

$\Phi_g^* H = \frac{1}{2}(q \ p) \underbrace{A_g^t A_g}_{= \mathbb{1}} \begin{pmatrix} q \\ p \end{pmatrix}$ $A_g \in SO(2)$ $A_g = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$

$\Phi_g(q,p) = (\cos\theta\, q + \sin\theta\, p, -\sin\theta\, q + \cos\theta\, p)$

does it preserve ω?

$\Phi_g^* \omega = d(\cos\theta\, q + \sin\theta\, p) \wedge d(-\sin\theta\, q + \cos\theta\, p)$

$= (\cos\theta\, dq + \sin\theta\, dp) \wedge (-\sin\theta\, dq + \cos\theta\, dp)$

$= \cos(\theta)^2 dq \wedge dp - \sin(\theta)^2 dp \wedge dq = dq \wedge dp = \omega$

conserved quantity is $q^2 + p^2$

harmonic oscillator in n-dimensions

$M = T^*\mathbb{R}^n \cong \mathbb{R}^{2n}$

coordinates (q,p) $\quad q \in \mathbb{R}^n \quad p \in \mathbb{R}^n$

$H(q,p) = \frac{1}{2}\left(\|q\|^2 + \|p\|^2\right) = \frac{1}{2}(q\ p)\begin{pmatrix} q \\ p \end{pmatrix}$

$\omega = dq^\mu \wedge dp_\mu = dq^\mu \otimes dp_\mu - dp_\mu \otimes dq^\mu$

$$\Phi_g : \begin{pmatrix} q \\ p \end{pmatrix} \in \mathbb{R}^{2n} \mapsto \overbrace{\begin{pmatrix} A & B \\ C & D \end{pmatrix}}^{N} \begin{pmatrix} q \\ p \end{pmatrix} \in \mathbb{R}^{2n}$$

preserving H: $\quad \frac{1}{2}(q\ p) N^t N \begin{pmatrix} q \\ p \end{pmatrix} = \frac{1}{2}(q\ p)\begin{pmatrix} q \\ p \end{pmatrix} \rightarrow N \in O(2n)$

can show that N preserves ω if $N^t \Omega N = \Omega = \begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix}$

$\omega = (dq\ dp)\begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix}\begin{pmatrix} dq \\ dp \end{pmatrix} \qquad d\left(N\begin{pmatrix} q \\ p \end{pmatrix}\right) = N\begin{pmatrix} dq \\ dp \end{pmatrix}$

$\phi_g^* \omega = (dq\ dp)\, N^t \begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix} N \begin{pmatrix} dq \\ dp \end{pmatrix}$

$\{ N \in GL(2n,\mathbb{R}) \mid N^t \Omega N = \Omega \} = Sp(2n,\mathbb{R})$ symplectic group

Φ symmetry if $N \in Sp(2n,\mathbb{R}) \cap O(2n)$

$$= \left\{ \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \middle| \ A^tA + B^tB = \mathbb{1} \quad A^tB = B^tA \right\}$$

$$\cong U(n)$$

$$\mathfrak{g} = \left\{ \begin{pmatrix} X & Y \\ -Y & X \end{pmatrix} \middle| \ X^t = -X, \ Y^t = Y \right\}$$

n^2 conserved quantities

in QFT we (can) do this

$\varphi : \mathbb{R}^4 \to \mathbb{R}$ scaler field

$\varphi : \mathbb{R} \times \mathbb{R}^3 \to \mathbb{R}$

treat φ as a map $\mathbb{R} \to C^\infty(\mathbb{R}^3, \mathbb{R})$

$f(t) = \varphi(t, \cdot) \to$ that is infinite collection of points

$$\varphi(t, \cdot) \equiv \{ \varphi(t, \vec{x}) \mid \vec{x} \in \mathbb{R}^3 \}$$

Klein-Gordon field $\partial_\mu \partial^\mu \varphi = 0 \to \partial_t \phi^2 - \nabla^2 \phi = 0$

$\to \dot{f}(t) = \nabla^2 f(t)$ $\nabla^2 : C^\infty(\mathbb{R}^3, \mathbb{R}) \to C^\infty(\mathbb{R}^3, \mathbb{R})$

<u>momentum</u> π $[\varphi(t, \vec{x}), \pi(t, \vec{x}')] = i\hbar\, \delta^{(3)}(x, \vec{x})$

<u>general approach</u> box of height L, Fourier transform

$[\hat{\varphi}_k(t), \hat{\pi}_{k'}(t)] = i\hbar\, \delta_{k,k'}$ $\hat{\varphi}_k$ and $\hat{\pi}_k$ in terms of Harmonic oscillators

$[q, p] = i\hbar$ $a \propto q + ip$ $[a, a^\dagger] = \mathbb{1}$

$a^\dagger \propto q - ip$

$\mathcal{L} = \frac{1}{2} \mathrm{Tr}(F_{\mu\nu} F^{\mu\nu})$

Electromagnetism fields are A_μ (4 potential)

$A_\mu + \partial_\mu \lambda$ is also the same field

physical fields are $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

$$\int D[A]\, e^{-iS(A)}$$